Ashutosh Sharma

Cloud Engineering · AWS · Terraform · Docker · Kubernetes · GitHub Actions · Python
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Summary

IT undergraduate who builds cloud systems on AWS as code and tests them by breaking them. LiftLens runs on Lambda, DynamoDB and Cognito with all 25 resources in Terraform, and its API and data are live at $0 a month. SwimSync is a FastAPI and PostgreSQL booking system on Kubernetes, fault-tested by killing instances, the database link and pods, and monitored with Prometheus and Grafana. Seeking a cloud engineering internship.

Education

Netaji Subhas University of Technology (NSUT) **Expected 2028**
B.Tech in Information Technology *New Delhi, India*

- Relevant Coursework: Data Structures and Algorithms, Advanced Programming (Java, grade **9/10**)
- LeetCode: **64** solved (33 Easy, 24 Medium, 7 Hard)

American International School of Mozambique (AISM) **2023**
Class XII, High School Diploma · 89% *Maputo, Mozambique*

Experience

Digital Marketing Intern **May 2025 – Jul 2025**
Socialbent *Gurgaon, Haryana*

- Researched markets, competitors and users for client campaigns, alongside SEO research and content optimization.
- Monitored campaign performance with the marketing team as part of campaign management.

Independent Researcher **Sep 2026 – Present**
Re-evaluation of spike-aware cloud CPU forecasting *New Delhi, India*

- Identified **two** flawed metrics in a 2025 *Future Generation Computer Systems* (FGCS) cloud CPU forecasting paper, confirmed in the authors' code; replicated it on the **29 GB** Alibaba 2018 cluster trace with Python (pandas), showing a simpler approach matches its spike recall with **14 to 29%** less over-provisioning across 3 seeds with no actuation delay; with a 15-minute delay, the paper's method wins.

Projects

LiftLens | *React, TypeScript, Python, AWS Lambda, DynamoDB, Cognito, Terraform*

- Built a 3-tier serverless web app (React/TypeScript, Python on AWS Lambda, DynamoDB, Cognito) managed entirely with Terraform (**25** resources across three stacks); the API and data are live, and app hosting on CloudFront is next.
- Implemented on-device pose estimation with MediaPipe so photos are never uploaded, and designed an explainable score with a **±2.4%** noise band calibrated from repeat photos.
- Achieved a **$0** monthly cost with DynamoDB on provisioned capacity inside the free tier; a **$1** budget alert, a kill-switch Lambda and a deploy lock, tested end to end, limit the damage if that changes.
- Implemented JWT verification against Cognito to secure the API, and covered it with **89** TypeScript and **112** Python tests, including forged-signature and expired-token cases.
- Automated builds and checks: a GitHub Actions CI pipeline runs lint, both test suites, the frontend build and Terraform validation on every pull request, and Bash scripts cross-build the Lambda package for Linux arm64; **8** ADRs.

SwimSync | *Python, FastAPI, PostgreSQL, React, Docker, Kubernetes, Terraform, Prometheus, Grafana*

- Eliminated requests dropped during Kubernetes rolling restarts by giving each pod a 5 s pause before stopping: **4 errors in 308** requests before, **none in 291** after, on a local kind cluster that CI builds and books through on every push.
- Built a booking system for capacity-limited swimming sessions, run with Docker Compose: a PostgreSQL row lock makes the capacity check and the booking one atomic step, idempotency keys make retries safe, a transactional outbox sends notifications, and an operator screen cuts capacity or closes sessions with an audit log.
- Validated it with **60** backend and **37** frontend tests and **three** fault drills (instance killed; database down, slow or losing a commit reply; pod restarts and kills), all with **zero** duplicate or false bookings; one exposed a worker crash, fixed so all **473** notifications arrived.
- Designed the AWS deployment in Terraform (two-zone VPC, ALB, RDS, IMDSv2, Session Manager via VPC endpoints), validated in CI and not deployed, at ₹0; recorded **18** decision records.
- Added Prometheus and Grafana as plain manifests after measuring the kube-prometheus-stack Helm chart overload the cluster (**48,642** series against **1,071**); the API's counters matched a **3,809**-attempt load test exactly.

Technical Skills

Cloud (AWS): Lambda, DynamoDB, Cognito, CloudWatch, SNS, Budgets, IAM and IAM Identity Center (SSO)
Infrastructure: Terraform (multi-stack layout, plan/apply review, deletion protection), Docker Compose, Kubernetes (kind), Prometheus, Grafana
Languages and Data: Python (pandas), TypeScript, JavaScript, Java, SQL, Bash, HTML, CSS, Excel
Web and Data: React, Vite, MediaPipe (in-browser pose detection), FastAPI, PostgreSQL
Practices: automated testing (Vitest, pytest), architecture decision records, Git and pull-request workflow, GitHub Actions CI

Certifications

NPTEL Cyber Security and Privacy (IIT Madras) — Elite + Gold, **93%**, top **1%** of 20,949 · AWS Certified Cloud Practitioner — in preparation · AWS Solutions Architect – Associate — planned